

ASTA607_22424447

!a) Describe the dataset

```
library(MASS)
```

```
data(crabs)
```

```
str(crabs)
```

```
summary(crabs)
```

Output:

```
> summary(crabs)
```

sp	sex	index	FL	RW
B:100	F:100	Min. : 1.0	Min. : 7.20	Min. : 6.50
O:100	M:100	1st Qu.:13.0	1st Qu.:12.90	1st Qu.:11.00
		Median :25.5	Median :15.55	Median :12.80
		Mean :25.5	Mean :15.58	Mean :12.74
		3rd Qu.:38.0	3rd Qu.:18.05	3rd Qu.:14.30
		Max. :50.0	Max. :23.10	Max. :20.20

CL	CW	BD
Min. :14.70	Min. :17.10	Min. : 6.10
1st Qu.:27.27	1st Qu.:31.50	1st Qu.:11.40
Median :32.10	Median :36.80	Median :13.90
Mean :32.11	Mean :36.41	Mean :14.03
3rd Qu.:37.23	3rd Qu.:42.00	3rd Qu.:16.60
Max. :47.60	Max. :54.60	Max. :21.60

1b)

```
# Load the MASS package
```

```
library(MASS)
```

```
# Load the crabs dataset
```

```
data(crabs)
```

```
# Display the structure of the dataset
```

```
str(crabs)
```

Output:

```
# Load the crabs dataset
```

```
> data(crabs)
```

```
>
```

```
> # Display the structure of the dataset
```

```
> str(crabs)
```

```
'data.frame': 200 obs. of 8 variables:
```

```
$ sp : Factor w/ 2 levels "B","O": 1 1 1 1 1 1 1 1 1 1 ...
```

```
$ sex : Factor w/ 2 levels "F","M": 2 2 2 2 2 2 2 2 2 2 ...
```

```
$ index: int 1 2 3 4 5 6 7 8 9 10 ...
```

```
$ FL : num 8.1 8.8 9.2 9.6 9.8 10.8 11.1 11.6 11.8 11.8 ...
```

```
$ RW : num 6.7 7.7 7.8 7.9 8 9 9.9 9.1 9.6 10.5 ...
```

```
$ CL : num 16.1 18.1 19 20.1 20.3 23 23.8 24.5 24.2 25.2 ...
```

```
$ CW : num 19 20.8 22.4 23.1 23 26.5 27.1 28.4 27.8 29.3 ...
```

```
$ BD : num 7 7.4 7.7 8.2 8.2 9.8 9.8 10.4 9.7 10.3 ...
```

```
>
```

1c)

1c)

Load MASS package

library(MASS)

Load dataset

data(crabs)

Frequency table for species

table(crabs\$sp)

Frequency table for sex

table(crabs\$sex)

Cross-tabulation (Species by Sex)

table(crabs\$sp, crabs\$sex)

OUTPUT:

```
> ### 1c)
> # Load MASS package
> library(MASS)
>
> # Load dataset
> data(crabs)
>
> # Frequency table for species
> table(crabs$sp)
```

```
  B    O
100 100
```

```
>
> # Frequency table for sex
> table(crabs$sex)
```

```
  F    M
100 100
```

```
>
> # Cross-tabulation (Species by Sex)
> table(crabs$sp, crabs$sex)
```

```
      F    M
B  50  50
O  50  50
```

1d.

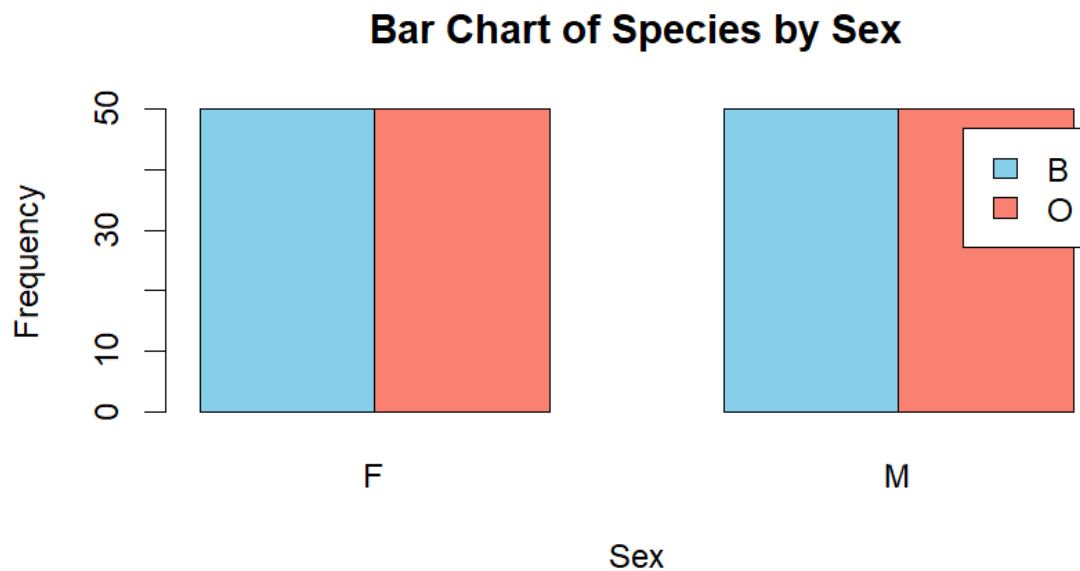
Bar plot

```

barplot(tab,
        beside = TRUE,
        col = c("skyblue", "salmon"),
        legend = rownames(tab),
        main = "Bar Chart of Species by Sex",
        xlab = "Sex",
        ylab = "Frequency")

```

OUTPUT:



1e.

The crabs dataset is perfectly balanced across the four species–sex groups, with 50 observations in each group. This results in a balanced 2×2 factorial design, which is ideal for comparative and multivariate statistical analysis.

Question 2

2a.

Load MASS package

library(MASS)

Load dataset

```
data(crabs)
```

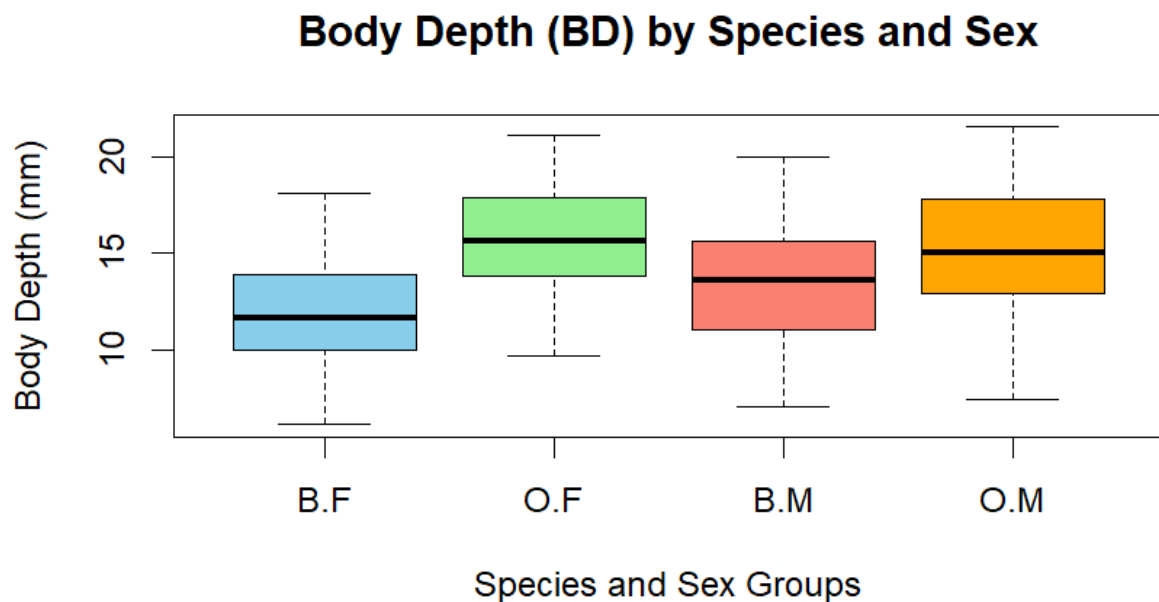
Create combined group variable

```
crabs$group <- interaction(crabs$sp, crabs$sex)
```

Side-by-side boxplots

```
boxplot(BD ~ group,  
  data = crabs,  
  col = c("skyblue", "lightgreen", "salmon", "orange"),  
  main = "Body Depth (BD) by Species and Sex",  
  xlab = "Species and Sex Groups",  
  ylab = "Body Depth (mm)")
```

OUTPUT:



2b.

The side-by-side boxplots show minimal evidence of extreme outliers across the four species–sex groups. The median body depth is higher for males than females in both species, indicating a strong sex effect. Differences between species are less pronounced than

differences between sexes. The interquartile ranges are similar across groups, although male groups exhibit slightly greater variability. Overall, body depth appears more strongly associated with sex than species.

Question 3

3a.

Load MASS package

library(MASS)

Load dataset

data(crabs)

Set plotting area for side-by-side plots

par(mfrow = c(1, 2))

Histogram for Blue species

```
hist(crabs$CL[crabs$sp == "B"],  
      main = "Histogram of CL (Blue Species)",  
      xlab = "Carapace Length (mm)",  
      col = "skyblue",  
      border = "black")
```

Histogram for Orange species

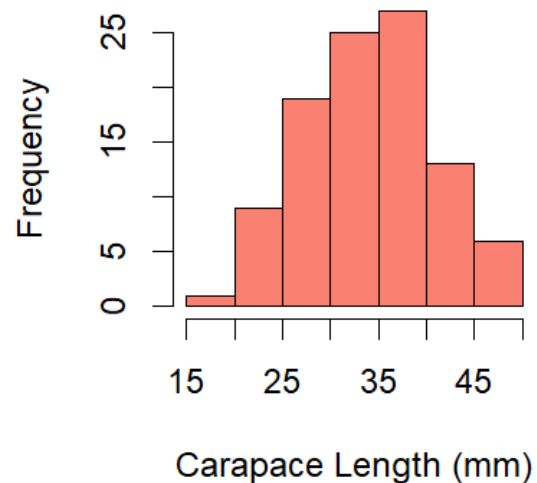
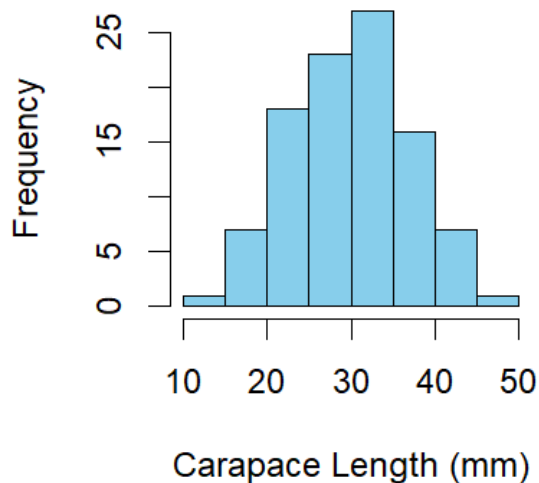
```
hist(crabs$CL[crabs$sp == "O"],  
      main = "Histogram of CL (Orange Species)",  
      xlab = "Carapace Length (mm)",  
      col = "salmon",  
      border = "black")
```

Reset plotting area

par(mfrow = c(1, 1))

OUTPUT:

Histogram of CL (Blue Specie Histogram of CL (Orange Spec



3b.

The histograms of carapace length for both Blue and Orange species are approximately symmetric and roughly bell-shaped, suggesting near-normal distributions. There is no strong evidence of skewness in either species. However, mild bimodality may be observed, which is likely attributable to the inclusion of both male and female crabs within each species. Overall, the shapes of the two species distributions are very similar.

QUESTION 4

4a.

```
library(dplyr)
```

```
crabs %>%
```

```
  group_by(sp, sex) %>%
```

```
  summarise(across(c(FL, RW, CL, CW, BD), mean), .groups = "drop")
```

output:

```
> crabs %>%
+   group_by(sp, sex) %>%
```

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```
+ summarise(across(c(FL, RW, CL, CW, BD), mean), .groups = "drop")
```

```
# A tibble: 4 × 7
```

	sp	sex	FL	RW	CL	CW	BD
	<fct>	<fct>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	B	F	13.3	12.1	28.1	32.6	11.8
2	B	M	14.8	11.7	32.0	36.8	13.4
3	O	F	17.6	14.8	34.6	39.0	15.6
4	O	M	16.6	12.3	33.7	37.2	15.3

4b.

```
# Load MASS package  
library(MASS)
```

```
# Load dataset  
data(crabs)
```

```
# Compute group means  
means <- aggregate(cbind(FL, RW, CL, CW, BD) ~ sp + sex,  
                   data = crabs,  
                   FUN = mean)
```

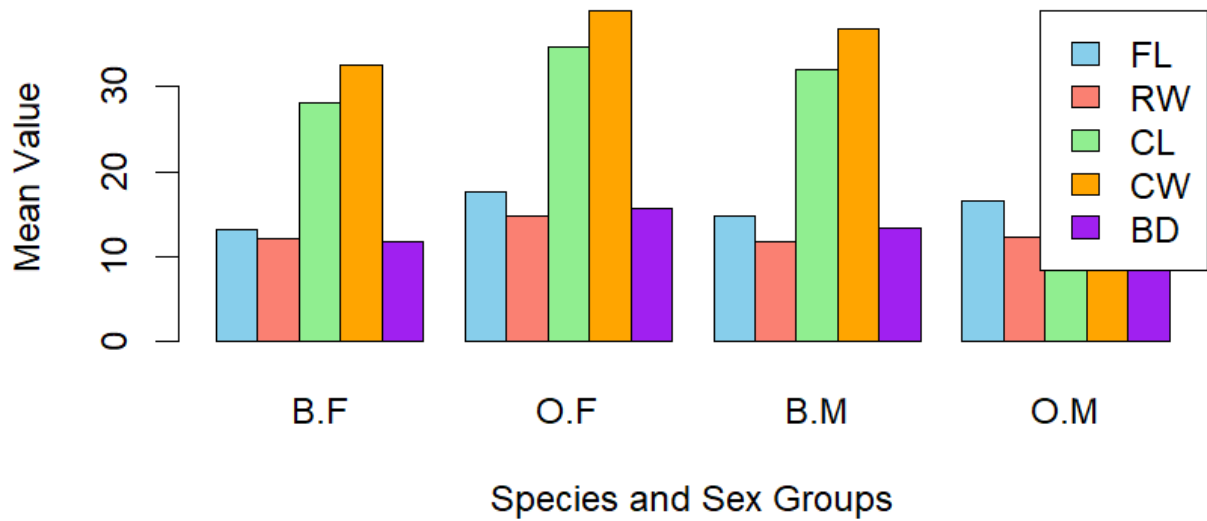
```
# Create group labels  
means$group <- paste(means$sp, means$sex, sep = ".")
```

```
# Create matrix of means (rows = measurements, columns = groups)  
mean_matrix <- t(as.matrix(means[, 3:7]))  
colnames(mean_matrix) <- means$group
```

```
# Grouped bar chart  
barplot(mean_matrix,  
        beside = TRUE,  
        col = c("skyblue", "salmon", "lightgreen", "orange", "purple"),  
        legend.text = rownames(mean_matrix),  
        args.legend = list(x = "topright"),  
        main = "Mean Morphological Measurements by Species and Sex",  
        xlab = "Species and Sex Groups",  
        ylab = "Mean Value")
```

OUTPUT:

Mean Morphological Measurements by Species and Sex



4c.

```
library(MASS)
```

```
library(dplyr)
```

```
data(crabs)
```

```
# Compute means by species and sex
```

```
means <- crabs %>%
```

```
  group_by(sp, sex) %>%
```

```
  summarise(across(c(FL, RW, CL, CW, BD), mean), .groups = "drop")
```

```
means
```

```
output:
```

```
# A tibble: 4 × 7
```

	sp	sex	FL	RW	CL	CW	BD
	<fct>	<fct>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	B	F	13.3	12.1	28.1	32.6	11.8
2	B	M	14.8	11.7	32.0	36.8	13.4
3	O	F	17.6	14.8	34.6	39.0	15.6
4	O	M	16.6	12.3	33.7	37.2	15.3

>

Species differences (averaging over sex)

```
species_means <- crabs %>%
```

```
  group_by(sp) %>%
```

```
  summarise(across(c(FL, RW, CL, CW, BD), mean))
```

```
species_diff <- abs(species_means[1, -1] - species_means[2, -1])
```

```
species_diff
```

output:

```
  FL    RW    CL    CW    BD
1 3.054 1.621 4.095 3.395 2.895
```

Sex difference(averaging over species)

```
sex_means <- crabs %>%
```

```
  group_by(sex) %>%
```

```
  summarise(across(c(FL, RW, CL, CW, BD), mean))
```

```
sex_diff <- abs(sex_means[1, -1] - sex_means[2, -1])
```

```
sex_diff
```

output:

```
FL    RW    CL    CW    BD
1 0.302 1.497 1.491 1.169 0.613
```

Answer:

Carapace Width (CW) exhibits the greatest difference between species, indicating it is most sensitive to species variation. Frontal Lobe (FL) shows the least variation between sexes, suggesting that sexual dimorphism has the smallest effect on this measurement compared to the others.

Question 5

5a.

Define colours by species

```
cols <- ifelse(crabs$sp == "B", "skyblue", "salmon")
```

```
pairs(measurements,
```

```
  main = "Scatterplot Matrix by Species",
```

pch = 19,

col = cols)

output:

